

AN ELEMENTARY CLASSIFICATION OF THE GROUPS OF ORDER 81

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Thesis

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ABSTRACT

The classification of finite, non-abelian groups is of interest, not only to algebraists, but to mathematicians in all fields. In general, little is known about non-abelian p -groups. However, cyclic extensions provide a convenient way to classify such groups. One particularly interesting set of p -groups is the groups of order p^4 . Much has been done to classify these groups, but little has been done using cyclic extensions to do so. In this paper, I will deal with the non-abelian groups of order 81. Along the way, many general results applicable for non-abelian groups of order p^4 will be presented. A first step in using cyclic extensions for such a problem is to prove the Cyclic Extension Theorem.

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CHAPTER I

INTRODUCTION

Since the development of modern abstract algebra, it has been of great interest to classify all groups of finite order. For instance, given a finite group G , one might ask if G is abelian, cyclic, expressed as a direct or semi-direct product, and so on. In addition, algebraists have tried to count the number of finite groups. As expected, this is not a trivial task. Often sophisticated techniques are necessary to classify groups of particular orders.

As it turns out, groups of prime-power order, called p -groups, demonstrate some unique and interesting properties. Ludvig Sylow did significant work in this field and his analysis gives mathematicians many useful facts about p -groups. Unfortunately, Sylow analysis doesn't always complete the whole picture. For instance, Sylow analysis is sufficient to classify all groups of order p^2q , where p and q are distinct primes, but is not sufficient to classify groups of order p^4 .

In most basic algebra texts, [1], for example, the classification of groups of orders from 2 through 15 is discussed. However, classifying groups of order $16 = 2^4$ is more complex and is usually omitted. Marcel Wild has recently produced an elementary classification in [2] of such groups. One of his main tools is what he called the Cyclic Extension Theorem. According to Wild, a somewhat disguised version of

the theorem originally appeared in early work done by Otto Hölder [3]. Within this publication, Hölder explored inner automorphisms and utilized direct and semi-direct products, along with a version of the Cyclic Extension Theorem, to classify groups of order p^3 , pq^2 , pqr and p^4 , for p, q, r distinct primes. At the conclusion of [2], Wild expresses interest in finding an elementary proof of the Cyclic Extension Theorem, one that does not use Artin-Schreier theory.

The main goals of this thesis are to provide an elementary proof of the Cyclic Extension Theorem and to use cyclic extensions to classify the ten non-abelian groups of order $81 = 3^4$. Although, up to isomorphism, there are also five abelian groups of order 81, the classification of the non-abelian groups of order 81 are those which are featured here. Certain techniques that are used for classifying groups of order 81 can be generalized to the classification of groups of order p^4 .

CHAPTER II

THE CYCLIC EXTENSION THEOREM

2.1 Cyclic Extensions

Suppose there is a finite group G . Consider a subgroup $N \triangleleft G$, where G/N is cyclic, i.e. $G/N \cong C_n$, the cyclic group of order n . Now, choose any $a \in G - N$ such that the coset Na generates G/N . Let $v = a^n \in N$. In addition, let $\tau \in \text{Aut}(N)$ act via conjugation by a . Thus,

$$\tau(v) = aa^na^{-1} = a^n = v.$$

In other words, τ fixes v . Also, for $x \in N$,

$$\tau^n(x) = a^nxa^{-n} = vxv^{-1}.$$

Thus, τ^n acts via conjugation by v . In particular, if N is an abelian group, then τ^n is the identity automorphism.

Definition 1 An *extension type* for a group N is any quadruple (N, n, τ, v) , satisfying the above conditions.

Notice that this definition is stated without mention of a group G . However, starting with a group G having a factor group $G/N \cong C_n$, one can construct an extension type (N, n, τ, v) as above. Moreover, the extension type determines G up to isomorphism.

2.2 Using Extension Types to Construct Groups

What guarantees that each extension type can be realized by a group G ?

Theorem 1 (The Cyclic Extension Theorem) Each extension type (N, n, τ, v) is realized by a suitable group G .

Until now it appears that no elementary proof of this theorem existed, since it often appears as a special case of Artin-Schreier theory. However, an elementary proof to the theorem is featured here.

Proof Let G be the set of ordered pairs (x, a^i) for $x \in N$ and $i \in \{0, 1, \dots, n-1\}$. In addition, G is coupled with a binary operation $*$ whose definition follows.

$$(x, a^i) * (y, a^j) = \left(x\tau^i(y)v^{\lfloor \frac{i+j}{n} \rfloor}, a^{i+j-n\lfloor \frac{i+j}{n} \rfloor} \right)$$

To ensure that $(G, *)$ is a group, it remains to show that $*$ is a group operation. That is, it suffices to show that $*$ satisfies the three axioms of a group operation. A first step in doing so is to notice that (e, a^0) is the identity in G since

$$(x, a^i) * (e, a^0) = (x\tau^i(e), a^{i+0}) = (x, a^i).$$

Also, $(x, a^i)^{-1} = (\tau^{-i}[(vx)^{-1}], a^{n-i})$ since

$$\begin{aligned} (x, a^i) * (\tau^{-i}[(vx)^{-1}], a^{n-i}) &= (x\tau^i\{\tau^{-i}[(vx)^{-1}]\}v, a^{i+n-i-n}) \\ &= (x(vx)^{-1}v, a^0) \\ &= (xx^{-1}v^{-1}v, a^0) \\ &= (e, a^0). \end{aligned}$$

Finally, we will show $*$ is associative. We will use the elementary fact that, for all integers m and r ,

$$v^r \tau^{m-nr}(z) = \tau^m(z) v^r.$$

Consider

$$\begin{aligned}
& [(x, a^i) * (y, a^j)] * (z, a^k) \\
&= \left(x\tau^i(y) v^{\lfloor \frac{i+j}{n} \rfloor}, a^{i+j-n\lfloor \frac{i+j}{n} \rfloor} \right) * (z, a^k) \\
&= \left(x\tau^i(y) v^{\lfloor \frac{i+j}{n} \rfloor} \tau^{i+j-n\lfloor \frac{i+j}{n} \rfloor}(z) v^{\lfloor \frac{i+j+k-n\lfloor \frac{i+j}{n} \rfloor}{n} \rfloor}, a^{i+j-n\lfloor \frac{i+j}{n} \rfloor + k - n\lfloor \frac{i+j+k-n\lfloor \frac{i+j}{n} \rfloor}{n} \rfloor} \right) \\
&= \left(x\tau^i(y) \tau^{i+j}(z) v^{\lfloor \frac{i+j}{n} \rfloor} v^{\lfloor \frac{i+j+k-n\lfloor \frac{i+j}{n} \rfloor}{n} \rfloor}, a^{i+j+k-n\lfloor \frac{i+j+k}{n} \rfloor} \right) \\
&= \left(x\tau^i(y) \tau^{i+j}(z) v^{\lfloor \frac{i+j+k}{n} \rfloor}, a^{i+j+k-n\lfloor \frac{i+j+k}{n} \rfloor} \right) \\
&= \left(x\tau^i[y\tau^j(z)] v^{\lfloor \frac{i+j+k}{n} \rfloor}, a^{i+j+k-n\lfloor \frac{i+j+k}{n} \rfloor} \right). \\
&(x, a^i) * [(y, a^j) * (z, a^k)] \\
&= (x, a^i) * \left(y\tau^j(z) v^{\lfloor \frac{j+k}{n} \rfloor}, a^{j+k-n\lfloor \frac{j+k}{n} \rfloor} \right) \\
&= \left(x\tau^i[y\tau^j(z) v^{\lfloor \frac{j+k}{n} \rfloor}] v^{\lfloor \frac{i+j+k-n\lfloor \frac{j+k}{n} \rfloor}{n} \rfloor}, a^{i+j-n\lfloor \frac{j+k}{n} \rfloor + k - n\lfloor \frac{i+j+k-n\lfloor \frac{j+k}{n} \rfloor}{n} \rfloor} \right) \\
&= \left(x\tau^i[y\tau^j(z)] v^{\lfloor \frac{j+k}{n} \rfloor} v^{\lfloor \frac{i+j+k-n\lfloor \frac{j+k}{n} \rfloor}{n} \rfloor}, a^{i+j+k-n\lfloor \frac{i+j+k}{n} \rfloor} \right) \\
&= \left(x\tau^i[y\tau^j(z)] v^{\lfloor \frac{i+j+k}{n} \rfloor}, a^{i+j+k-n\lfloor \frac{i+j+k}{n} \rfloor} \right).
\end{aligned}$$

Hence, $*$ is a group operation and the Cyclic Extension Theorem is proven.

Now that G is established to be a group, we will write a for (e, a) and x for (x, a^0)

2.3 Equivalence of Extension Types

In order to construct all groups of order 81, it would be sufficient to construct all extension types $(N, 3, \tau, v)$, where N is a group of order 27. However, the number of such extension types is large. We will present some techniques for identifying when two extension types realize isomorphic groups.

Definition 2 Two extension types are *equivalent* if they realize isomorphic groups.

Note that in [2], equivalence has the following meaning.

Definition 3 The extension types (N, n, τ, v) and (N', n, σ, ω) are *conjugate* if there is an isomorphism $\phi : N \rightarrow N'$ such that $\sigma = \phi \circ \tau \circ \phi^{-1}$ and $\omega = \phi(v)$.

Lemma 1 Conjugate extension types are equivalent.

Proof The proof appears in [2].

Lemma 2 Let N be abelian of odd order. Then, the extension types $(N, 3, \tau, v)$ and $(N, 3, \tau, v^2)$ are conjugate.

Proof Define an isomorphism $\phi : N \rightarrow N$ by $\phi(x) = x^2$. Then,

$$\phi(\tau(x)) = (\tau(x))^2 = \tau(x^2) = \tau(\phi(x))$$

$$\Rightarrow \phi\tau = \tau\phi$$

$$\Rightarrow \phi\tau\phi^{-1} = \tau$$

Since $\phi\tau\phi^{-1} = \tau$ and $\phi(v) = v^2$, the extension types $(N, 3, \tau, v)$ and $(N, 3, \tau, v^2)$ are conjugate.

Definition 4 Define a function $\mathcal{N}_\tau : N \rightarrow N$ by

$$\mathcal{N}_\tau(x) = x\tau(x)\tau^2(x) \cdots \tau^{n-1}(x).$$

Lemma 3 Let N be abelian. Then, for $x \in N$, the extension types (N, n, τ, v) and $(N, n, \tau, \mathcal{N}_\tau(x)v)$ are equivalent.

Proof Let G be a group realized by (N, n, τ, v) and constructed as in Section 2.2. We will show that G is also realized by $(N, n, \tau, \mathcal{N}_\tau(x)v)$. Let $a' = xa$. Then, define τ' to be the automorphism that acts via conjugation by a' . For $y \in N$, consider

$$\tau'(y) = a'y(a')^{-1} = (xa)y(a^{-1}x^{-1}) = x(aya^{-1})x^{-1} = x\tau(y)x^{-1} = \tau(y),$$

since N is abelian. Hence, $\tau' = \tau$. Then, define

$$\begin{aligned} v' &= (a')^n \\ &= (xa)^n \\ &= (xa)(xa) \cdots (xa) \\ &= (xa)(xa^{-1}a^2)(xa^{-2}a^3) \cdots (xa^{-(n-1)}a^n) \\ &= x(axa^{-1})(a^2xa^{-2}) \cdots (a^{n-1}xa^{-(n-1)})a^n \\ &= x\tau(x)\tau^2(x) \cdots \tau^{n-1}(x)v \\ &= \mathcal{N}_\tau(x)v. \end{aligned}$$

Thus, G is realized by $(N, n, \tau, \mathcal{N}_\tau(x)v)$.

Lemma 4 Let p be a prime. Let N be abelian. Then, the extension types (N, p, τ, v) and (N, p, τ^i, v^i) , for any $0 < i < p$, realize isomorphic groups.

Proof Let G be a group realized by (N, p, τ, v) . This extension type is determined by choices of N and $a \in G - N$. One could just as easily choose $a^i \in G - N$ and the resulting extension type would be (N, p, τ^i, v^i) . Thus, G is also realized by (N, p, τ^i, v^i) .

CHAPTER III

THE CONSTRUCTION OF ALL GROUPS OF ORDER 81

While the methods presented in this paper are adequate for classifying the 5 abelian groups of order 81, we will not consider these cases since they are well known and easily obtainable. In order to construct the non-abelian groups of order 81, we will construct, up to equivalence, all extension types $(N, 3, \tau, v)$, where N is a group of order 27.

3.1 Subgroups N

A corollary of Sylow's Theorem ensures that a group of order 81 has a normal subgroup N of order 27. There are five groups of order 27, three of which are abelian and two of which are non-abelian. When constructing extension types, the choice of N must come from the list of these five subgroups. It is desirable to cut this list down.

Theorem 2 Every group G of order p^4 has an abelian subgroup of order at least p^3 .

Proof Let Z denote the center of G . If $|Z| \geq p^3$, then we are done. Since p -groups have non-trivial centers, we may assume that $|Z| = p$ or $|Z| = p^2$. We first claim that G has a normal subgroup H of order p^2 . If $|Z| = p^2$, let $H = Z$.

Suppose from now on that $|Z| = p$. By Lemma 4.1.3 in [1], it is enough to find a normal subgroup of G/Z of order p . If G/Z is abelian, this is trivial. If G/Z is non-abelian, then the center of G/Z is a normal subgroup of G/Z , and the claim is proved. Now, define a homomorphism $\Phi : G \rightarrow \text{Aut}(H)$ by

$$[\Phi(g)](h) = ghg^{-1}.$$

If $x, h \in H$, then

$$\Phi(x)(h) = xhx^{-1} = h.$$

This implies, by definition, $H \subseteq \ker(\Phi)$.

We wish to show that this containment is strict. Suppose for a contradiction that $\ker(\Phi) = H$. Then, $|\ker(\Phi)| = |H| = p^2$, which gives $|G/\ker(\Phi)| = p^2$. By the First Isomorphism Theorem in [1], Φ corresponds to a one-to-one map from $G/\ker(\Phi)$ to $\text{Aut}(H)$. Therefore, $|G/\ker(\Phi)|$ must divide $|\text{Aut}(H)|$. However, notice that since $H = \ker(\Phi)$ is isomorphic to either C_{p^2} or $C_p \times C_p$ and $|\text{Aut}(H)| = (p^2 - p)$ or $|\text{Aut}(H)| = (p^2 - 1)(p^2 - p)$, respectively, neither of which is divisible by p^2 , we arrive at a contradiction.

Thus, there must exist at least one $g \in G - H$ such that g commutes with all elements of H . Since H is abelian and g commutes with all elements of H , the subgroup generated by H and g will be abelian. This subgroup has order at least p^3 . This establishes the theorem.

Now it is justified to focus only on the 3 abelian subgroups of order 27. However, a much stronger result holds. That is, if G contains C_{27} as a subgroup, it

also contains either $C_9 \times C_3$ or $C_3 \times C_3 \times C_3$. This allows for C_{27} to be removed from the relevant list of normal subgroups to consider when constructing extension types.

Theorem 3 If a nonabelian group G of order 81 contains a subgroup isomorphic to C_{27} , then G also contains a subgroup isomorphic to $C_9 \times C_3$.

Proof Let $H = \langle h \rangle$ be a cyclic subgroup of G of order 27. Choose $a \in G - H$, define $v = a^3$ and let $\tau \in \text{Aut}(H)$ act via conjugation by a . Then, by definition, $G = \langle h, a \rangle$. It remains to show that G contains a subgroup isomorphic to $C_9 \times C_3$. Since H is cyclic and has order 27, every automorphism of H will be of the form $x \mapsto x^m$, where m and 27 are relatively prime. Thus, $\text{Aut}(H)$ is cyclic of order 18 and has two elements of order 3. By Lemma 2, we only need to consider one of them. It's easily verified that $\tau(x) = x^{10}$ is such an element. So, assume without loss of generality that $\tau(x) = x^{10}$. Let H' be the subgroup of H generated by h^3 . This is a cyclic subgroup of order 9 and commutes with a since τ fixes h^3 . To see this, first notice that

$$\tau(h^3) = h^{30} = h^3.$$

Also, $\tau(h^3) = ah^3a^{-1}$, which gives that $ah^3 = h^3a$. Thus, it suffices to show the existence of an element $x \in G$ such that

$$|x| = 3$$

$$x \notin H'$$

$$xh^3 = h^3x.$$

Then, $\langle h^3, x \rangle \cong C_9 \times C_3$. Let $a' = ah^r$, where r is to be determined. Clearly, $ah^r \in G - H$. So, for $n \in H$, consider

$$\begin{aligned}
a'n(a')^{-1} &= (ah^r)n(h^{-r}a^{-1}) \\
&= a(h^rnh^{-r})a^{-1} \\
&= a(nh^rh^{-r})a^{-1} \\
&= ana^{-1} \\
&= \tau(n).
\end{aligned}$$

Thus, a' commutes with H' , just as a does. Lastly, it suffices to show that we can choose r such that a' has order 3. From the proof of Lemma 3,

$$(a')^3 = \mathcal{N}_\tau(h^r)v.$$

Thus, we must show that

$$\mathcal{N}_\tau(h^r) = v^{-1}$$

for an appropriate choice of r . Consider

$$\mathcal{N}_\tau(h^r) = h^r\tau(h^r)\tau^2(h^r) = h^rh^{10r}h^{100r} = h^{111r} = h^{3r}.$$

Thus, we must show that $v = h^{-3r}$ for some r . Equivalently, $v \in H'$. However, v is fixed by τ , and H' is the set of all fixed points of τ . This concludes the proof.

Together, Theorems 2 and 3 produce a powerful result. That is, any non-abelian group of order 81 contains a subgroup isomorphic to either $C_9 \times C_3$ or $C_3 \times C_3 \times C_3$. In particular, it is sufficient to consider only two choices of N when constructing

extension types. Conveniently, much is known about the automorphism groups of both $C_9 \times C_3$ and $C_3 \times C_3 \times C_3$.

3.2 Automorphisms τ of N

For $N = C_9 \times C_3$ and $N = C_3 \times C_3 \times C_3$, we must find enough automorphisms τ of N to construct, up to equivalence, all extension types of the form $(N, 3, \tau, v)$. The following lemma will be useful.

Lemma 5 Let $\mathcal{F}_p$ be the field with p elements and let m be a positive integer. Let $A \in GL_m(\mathcal{F}_p)$ have order p . Then, A is conjugate to a matrix in Jordan canonical form.

Proof Notice that A has the characteristic polynomial $(X - 1)^p$, which splits over $\mathcal{F}_p$. Thus, by Theorem 23 in [4], A is conjugate to a matrix in Jordan canonical form. The proof is complete.

Let's first assume $N = C_9 \times C_3$ and let x be a generator of C_9 and y be a generator of C_3 . Thus, each element of N has the form $x^u y^v$ for $0 \leq u \leq 8$ and $0 \leq v \leq 2$. So, every automorphism of $C_9 \times C_3$ has the form

$$\begin{cases} x \mapsto x^a y^b \\ y \mapsto x^c y^d \end{cases},$$

for a, c integers modulo 9 and b, d integers modulo 3. This automorphism can conveniently be represented as the matrix $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$. Note that composition of automorphisms is compatible with matrix multiplication. Also, by Lemma 1, we only need

to consider automorphisms up to conjugacy in $\text{Aut}(C_9 \times C_3)$. Consider the homomorphism from $\text{Aut}(C_9 \times C_3)$ into $GL_2(\mathcal{F}_p)$ which maps the matrix representation of an automorphism in $\text{Aut}(C_9 \times C_3)$ to a matrix in $GL_2(\mathcal{F}_p)$ by reducing the top row modulo 3. By Lemma 5, the image in $GL_2(\mathcal{F}_p)$ of an automorphism of order 3 in $\text{Aut}(C_9 \times C_3)$ is conjugate to either $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ or $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Therefore, we may assume that our matrix has either the form

$$\begin{bmatrix} \alpha & \beta \\ 0 & 1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} \alpha & \beta \\ 1 & 1 \end{bmatrix},$$

where $\alpha \equiv 1 \pmod{3}$ and $\beta \equiv 0 \pmod{3}$. There are 18 such matrices, since there are 3 choices for α and β in each case. However, some of these matrices are conjugate. Since each automorphism is represented by a matrix, this computation is fairly straightforward.

Lemma 6 If $N = C_9 \times C_3$, then, up to conjugacy and squaring, every automorphism of N of order 3 in matrix form is listed below.

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 6 \\ 1 & 1 \end{bmatrix}$$

Proof Consider the following calculations.

$$\begin{aligned} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}^{-1} &= \begin{bmatrix} \alpha + 3 & \beta \\ 1 & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} &= \begin{bmatrix} \alpha - \beta & \beta \\ 0 & 1 \end{bmatrix} \end{aligned}$$

These calculations alone significantly reduce the list of relevant automorphisms. Moreover, applying Lemma 4 further reduces the list to prove the lemma.

Next, assume $N = C_3 \times C_3 \times C_3$. It's well known that $C_3 \times C_3 \times C_3$ can be viewed as a three-dimensional vector space over C_3 . So, by Proposition 17 in [4], $\text{Aut}(C_3 \times C_3 \times C_3) \cong GL_3(C_3)$.

Lemma 7 Every automorphism of order 3 in $GL_3(\mathcal{F}_p)$ in matrix form is conjugate to one of the following two matrices.

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Proof This follows from Lemma 5.

3.3 Choices for v

The fourth component of an extension type is the element v . Since v is fixed by τ , it is necessary to find the fixed points of τ . Since each relevant automorphism is represented by a matrix, this is a matter of solving matrix equations. Then, Lemmas 2, 3 and 4 are applied to further reduce the list of elements v we must consider. There are seven cases corresponding to the seven automorphisms identified in the previous two lemmas.

1. For $\tau = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$, we have

$$\begin{aligned}
(\tau - I)(\bar{x}) = \bar{0} &\Leftrightarrow \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
&\Leftrightarrow \begin{pmatrix} 3b \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
&\Leftrightarrow 3b \equiv 0 \pmod{9} \\
&\Leftrightarrow b = 0.
\end{aligned}$$

So, the set of fixed points of τ is

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle.$$

Also, by Definition 4,

$$\mathcal{N}_\tau(x) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}^2 = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$$

which gives that $\text{image}(\mathcal{N}_\tau) = \left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\rangle$. First, applying Lemma 3 gives that

extension types containing $v = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$ and $v = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$ will be realized by isomorphic groups. Then, applying Lemma 2 gives that, for example, the extension types containing $v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ will be realized by isomorphic groups

since

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

So, by calculation, the only choices of v we must consider are

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}.$$

2. For $\tau = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$, we have

$$\begin{aligned} (\tau - I)(\bar{x}) = \bar{0} &\Leftrightarrow \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &\Leftrightarrow \begin{pmatrix} 3a \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &\Leftrightarrow 3a \equiv 0 \pmod{9} \\ &\Leftrightarrow a \equiv 0 \pmod{3} \\ &\Leftrightarrow a = 0, 3, 6. \end{aligned}$$

So, the set of fixed points of τ is

$$\left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle$$

Also, $\text{image}(\mathcal{N}_\tau) = \left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\rangle$. So, by calculation, the only choices of v we must consider are

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}.$$

3. For $\tau = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, we have

$$\begin{aligned} (\tau - I)(\bar{x}) = \bar{0} &\Leftrightarrow \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &\Leftrightarrow \begin{pmatrix} 0 \\ a \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ &\Leftrightarrow a \equiv 0 \pmod{3} \\ &\Leftrightarrow a = 0, 3, 6. \end{aligned}$$

So, the set of fixed points of τ is

$$\left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle.$$

Also, $\text{image}(\mathcal{N}_\tau) = \left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\rangle$. So, the only choices of v we must consider are

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}.$$

4. For $\tau = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$, we have

$$\begin{aligned}
 (\tau - I)(\bar{x}) = \bar{0} &\Leftrightarrow \begin{bmatrix} 0 & 3 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
 &\Leftrightarrow \begin{pmatrix} 3b \\ a \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
 &\Leftrightarrow 3b \equiv 0 \pmod{9} \text{ and } a \equiv 0 \pmod{3} \\
 &\Leftrightarrow a = 0, 3, 6 \text{ and } b = 0.
 \end{aligned}$$

So, the set of fixed points of τ is

$$\left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\rangle.$$

Also, $\text{image}(\mathcal{N}_\tau) = \left\langle \begin{pmatrix} 6 \\ 0 \end{pmatrix} \right\rangle$. So, by calculation, the only choice of v we must consider is

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

5. For $\tau = \begin{bmatrix} 1 & 6 \\ 1 & 1 \end{bmatrix}$, we have

$$\begin{aligned}
(\tau - I)(\bar{x}) = \bar{0} &\Leftrightarrow \begin{bmatrix} 0 & 6 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
&\Leftrightarrow \begin{pmatrix} 6b \\ a \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
&\Leftrightarrow 6b \equiv 0 \pmod{9} \text{ and } a \equiv 0 \pmod{3} \\
&\Leftrightarrow a = 0, 3, 6 \text{ and } b = 0.
\end{aligned}$$

So, the set of fixed points of τ is

$$\left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\rangle.$$

In this case, $\text{image}(\mathcal{N}_\tau) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, which contributes nothing useful. So, the only choices of v we must consider are

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\}.$$

6. For $\tau = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, we claim that for any non-trivial v , the extension type $(N, 3, \tau, v)$ is equivalent to one that we have already considered. To see this,

let G be a group realized by $(N, 3, \tau, v)$. It is enough to show that G contains a subgroup isomorphic to $C_9 \times C_3$. To do this, it is enough to find commuting elements $x, y \in G$ such that x has order 9, y has order 3 and $y \notin \langle x \rangle$.

Let $a \in G$ be an element such that τ acts via conjugation by a . Since v has order 3, a has order 9. The set S of fixed points of τ can be viewed as a 2-dimensional vector space with basis $\{(1, 0, 0), (0, 0, 1)\}$. Meanwhile, $\langle v \rangle$ is a 1-dimensional subspace of S . Pick $x = a$ and $y \in S - \langle v \rangle$.

Thus, the only choice of v we must consider is

$$\left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}.$$

7. For $\tau = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, we have

$$(\tau - I)(\vec{x}) = \vec{0} \Leftrightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} b \\ c \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Leftrightarrow b \equiv 0 \pmod{3} \text{ and } c \equiv 0 \pmod{3}$$

$$\Leftrightarrow b = c = 0.$$

So, the set of fixed points of τ is

$$\left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle.$$

Also, $\text{image}(\mathcal{N}_\tau) = \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle$. So, the only choice of v we must consider is

$$\left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}.$$

CHAPTER IV

A CLASSIFICATION

4.1 Extension Types of G

Now that all components of the extension types have been determined, it is now possible to list the extension types sufficient for constructing the nonabelian groups of order 81. For $N = C_9 \times C_3$, we have

$$\begin{aligned} & \left(N, 3, \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \\ & \left(N, 3, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right), \\ & \left(N, 3, \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 6 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 6 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right). \end{aligned}$$

For $N = C_3 \times C_3 \times C_3$, we have

$$\left(N, 3, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right).$$

4.2 The Isomorphism Problem

It remains only to find all equivalences among our 11 extension types. This is the “Isomorphism Problem” as described in [2].

In the following lemmas, N is assumed to be $C_9 \times C_3$.

Lemma 8 The following extension types are not equivalent.

$$\left(N, 3, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right)$$

Proof Let G_1 be realized by the extension type $\left(N, 3, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right)$ and G_2 be realized by the extension type $\left(N, 3, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right)$. Define $H = \{g^3 : g \in G\}$, for each group G_1 and G_2 . It follows that H is a normal subgroup of both G_1 and G_2 since it is a subgroup of the center of both G_1 and G_2 . Moreover,

$$H = \text{Image}(\mathcal{N}_\tau) = \left\langle \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\rangle.$$

First, consider G_1/H . By Lemma 5, since the image of $\tau = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$ in $GL_2(\mathcal{F}_p)$

is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, τ acts trivially on G_1/H . Hence, G_1/H is abelian. However, G_2/H

is non-abelian since the image of $\tau = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ in $GL_2(\mathcal{F}_p)$ is $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, which does not act trivially on G_2/H . This proves the lemma.

Lemma 9 The following extension types are equivalent.

$$\left(N, 3, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

Proof Assume that G is realized by the extension type

$$\left(N, 3, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

and construct G as in the proof of Theorem 1. Let $\langle x \rangle = C_9$ and $\langle y \rangle = C_3$. To construct a subgroup N' which is isomorphic to N , let $x' = a$ and $y' = x^6$. Then, $N' = \langle x', y' \rangle$. Define $a' = x^2$ so that $(a')^3 = v' = y'$. Also, let $\tau' \in \text{Aut}(N')$ be the automorphism that acts via conjugation by a' . Then, consider

$$\tau'(x') = a'x'(a')^{-1} = x^2ax^7 = x^2\tau(x^7)a = x^3a = x'y'.$$

$$\tau'(y') = a'y'(a')^{-1} = x^2x^6x^7 = x^6 = y'.$$

With respect to the generators x' and y' , τ' has matrix representation $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$.

Also, $v' = (a')^3$ has vector representation $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Consequently, G also realizes the extension type

$$\left(N, 3, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right).$$

This proves the lemma.

Theorem 4 Every non-abelian group of order 81 can be realized by precisely

one of the following extension types.

$$\begin{aligned} & \left(N, 3, \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \\ & \left(N, 3, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \\ & \left(N, 3, \begin{bmatrix} 1 & 6 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 6 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right), \\ & \left(N, 3, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right), \left(N, 3, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right) \end{aligned}$$

Proof We use the facts that two groups with non-isomorphic centers or two groups with a different number of elements of order 3 cannot be isomorphic. We will consider the center of each group and count the elements of order 3. The center of each group is the set of fixed points of τ . Counting the elements of order 3 is also straightforward. By Definition 1, G is the union of the three cosets N , Na and Na^2 . The number of elements of order 3 in N is easy to compute. In $N = C_9 \times C_3$, there are 8 elements of order 3. In $N = C_3 \times C_3 \times C_3$, there are 26 elements of order 3. So, for each N , it's enough to count the elements of order 3 in the coset Na , since the coset Na^2 will have an equal number of elements of order 3. For $xa \in Na$, by

the proof of Lemma 3, we know

$$(xa)^3 = \mathcal{N}_\tau(x)v.$$

Then, if $v \in \text{Image}(\mathcal{N}_\tau)$, the number of elements of order 3 in G is

$$(\text{The number of elements of order 3 in } N) + 2 |\ker(\mathcal{N}_\tau)|$$

Otherwise, the number of elements of order 3 in G is the same as the number of elements of order 3 in N . This information is summarized by the following tables.

τ, v	$Z(G)$	$v \in \text{Image}(\mathcal{N}_\tau)?$	$ \ker(\mathcal{N}_\tau) $	# of Elts. of o(3)
$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$	$C_3 \times C_3$	Yes	27	80
$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$	C_3	Yes	9	44

τ, v	$Z(G)$	$v \in \text{Image}(\mathcal{N}_\tau)?$	$ \ker(\mathcal{N}_\tau) $	# of Elts. of $\text{o}(3)$
$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}$	C_9	Yes	9	26
$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}$	C_9	No	9	8
$\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}$	$C_3 \times C_3$	Yes	9	26
$\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$C_3 \times C_3$	No	9	8
$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}$	$C_3 \times C_3$	Yes	9	26
$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$C_3 \times C_3$	No	9	8
$\begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}$	C_3	Yes	9	26
$\begin{bmatrix} 1 & 6 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix}$	C_3	Yes	27	62
$\begin{bmatrix} 1 & 6 \\ 1 & 1 \end{bmatrix}, \begin{pmatrix} 3 \\ 0 \end{pmatrix}$	C_3	No	27	8

From inspection of the above tables, it remains to apply Lemmas 8 and 9 to establish the theorem.

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